

- Q.18 What procedure should be preferred to analyze the impurities in water and steam?
- Q.19 Write a discussion about the dissolved oxygen analyzer.
- Q.20 What is the purpose of control loop in boiler?
- Q.21 Explain the controlling process of a super heater.
- Q.22 What is the need of air temperature determination during boiling operation?

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

- Q.23 Explain the air/fuel ratio control, furnace draft control and drum level control along with their uses in power generation.
- Q.24 What is the main purpose of a boiler, explain its full process.
- Q.25 Explain the process of power generation in thermal plant with neat block diagram.

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4th Sem / Instrumentation & Control

Subject : Power Plant Instrumentation

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

- Q.1 The alternator is used in power plants which converts
- a) Electrical Energy into Mechanical Energy
 - b) Electrical energy into Solar Energy
 - c) Mechanical Energy into Electrical Energy
 - d) Mechanical Energy into Nuclear Energy
- Q.2 The commercial sources of energy are
- a) Solar, wind and biomass
 - b) Fossil fuels, hydropower and nuclear energy
 - c) Wood, animal wastes and agriculture wastes
 - d) None of these

- Q.3 The main source of production of biogas is
- a) Human waste b) Wet cow dung
 - c) Wet livestock waste d) All above
- Q.4 In fuel cell, the chemical energy is converted into
- a) Mechanical b) Electrical
 - c) Heat d) Sound
- Q.5 The overall efficiency of thermal power plant is equal to
- a) Rankine cycle efficiency
 - b) Carnot cycle efficiency
 - c) Regenerative cycle efficiency
 - d) Boiler cycle efficiency
- Q.6 The area under the load curve represents
- a) The average load on the power system
 - b) Maximum demand
 - c) Number of units generated
 - d) Load factors

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 What is the purpose of conductivity cell?
- Q.8 What is spectrum analyzer?
- Q.9 What is the use of PH meter.
- Q.10 What is power factor?
- Q.11 What do you understand by solar power plant?
- Q.12 What is trivector meter?

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 What is the importance of instrumentation for power generation?
- Q.14 How frequency and power parameters are measured in power plants explain with diagram?
- Q.15 What are non-electrical parameters and how they are useful for any measurement process?
- Q.16 How temperature can be maintained through flow process?
- Q.17 Discuss the process to measure the smoke density in power plants.